

Writing and Solving Equations in Real-World Contexts

Worksheet 7A.01.07 **ANSWER KEY**

For questions 1 through 4, define the variable in a sentence, write an equation, solve it, and say what the solution means.

1. Peyton was given a \$40 gift card to her local coffee shop. She went to the shop four times and purchased the same thing each time. She has \$7.08 left on the card. What was the cost of each purchase?

Variable. $c = \text{the cost of one purchase.}$

Equation. $40 - 4c = 7.08$

$$-4c = -32.92$$

$$c = 8.23$$

Check: $40 - 4(8.23) = 40 - 32.92 = 7.08 \checkmark$

Meaning. *Peyton spent \$8.23 each time she went to the coffee shop.*

The card total goes down, so the rate is negative. $4c - 40 = 7.08$ is the common wrong start.

2. Georgina was given a \$50 gift card to a music app. She bought 18 songs that all cost the same amount. She has \$18.50 left on the card. How much did each song cost?

Variable. $s = \text{the cost of one song.}$

Equation. $50 - 18s = 18.50$

$$-18s = -31.50$$

$$s = 1.75$$

Check: $50 - 18(1.75) = 50 - 31.50 = 18.50 \checkmark$

Meaning. *Each song cost \$1.75.*

3. Shirley gives each of her eight grandchildren a \$15 gift card and the same amount of cash for their birthday. She spends \$280 in all. How much cash does she give each grandchild?

Variable. $c = \text{the amount of cash each grandchild receives.}$

Equation. $8(15 + c) = 280$

$$15 + c = 35$$

$$c = 20$$

Check: $8(15 + 20) = 8(35) = 280 \checkmark$

Meaning. *Shirley gives each grandchild \$20 in cash.*

Dividing both sides by 8 first is quicker than distributing. Every grandchild gets the same package, so the group repeats and the parentheses belong there.

4. Fatima ran 7 fewer miles this week than twice the number she ran last week. This week she ran 9 times as far as Luna, who ran 3 miles. How many miles did Fatima run last week?

Variable. $m = \text{the number of miles Fatima ran last week.}$

Equation. $2m - 7 = 27$

Luna ran 3 miles, so Fatima ran $9(3) = 27$ miles this week.

$$2m = 34$$

$$m = 17$$

Check: $2(17) - 7 = 34 - 7 = 27 \checkmark$

Meaning. *Fatima ran 17 miles last week.*

Work out the 27 before writing the equation. Students who see “7 fewer” and write $7 - 2m$ have the subtraction backwards, and they will not be able to explain their answer.

For questions 5 and 6, the coefficient is a fraction. Multiplying by the reciprocal is usually the quickest route.

5. Kiara is baking cookies for a bake sale. Each batch needs $1\frac{2}{3}$ cups of sugar. She has 12 cups, and her mother asked her to set aside $\frac{1}{3}$ cup for her coffee. How many batches can Kiara make?

Variable. $b = \text{the number of batches of cookies Kiara can make.}$

Equation. $1\frac{2}{3}b + \frac{1}{3} = 12 \quad (1\frac{2}{3} = 5/3)$

Set aside the $\frac{1}{3}$ cup first: $(5/3)b = 11\frac{2}{3}$

Multiply by the reciprocal: $b = (35/3) \times (3/5) = 7$

Check: $1\frac{2}{3}(7) + \frac{1}{3} = 11\frac{2}{3} + \frac{1}{3} = 12 \checkmark$

Meaning. *Kiara can make 7 batches of cookies.*

$11\frac{2}{3}$ written as an improper fraction is $35/3$. Multiplying by $3/5$ undoes multiplying by $5/3$.

6. A craft kit uses $\frac{1}{6}$ yard of ribbon for each bracelet, plus 2.5 yards for the display board. The kit contains 9 yards of ribbon in all. How many bracelets does the kit make?

Variable. $n = \text{the number of bracelets the kit makes.}$

Equation. $\frac{1}{6}n + 2.5 = 9$

$$\frac{1}{6}n = 6.5$$

Multiply both sides by 6: $n = 39$

Check: $39 \div 6 = 6.5$, and $6.5 + 2.5 = 9 \checkmark$

Meaning. *The kit makes 39 bracelets.*

Multiplying by $1/6$ is the same as dividing by 6, so the inverse is multiplying by 6 — not dividing by 6 again.

For questions 7 and 8, decide whether your answer makes sense in the situation before you write your sentence.

7. A charter bus for a field trip costs \$340 plus \$28 for each student who rides. The budget for the trip is \$1,000. How many students can go?

Variable. $s = \text{the number of students who can ride the bus.}$

Equation. $340 + 28s = 1,000$

$$28s = 660$$

$$s = 23.571\dots$$

23 students can go.

Check: $340 + 28(23) = 340 + 644 = \984 , and 24 students would cost \$1,012.

Meaning. *The trip can take 23 students and stay inside the \$1,000 budget.*

The equation gives a decimal and the situation does not allow one. Round DOWN here — 24 students goes over budget. Writing 23.571 students is the error to catch.

8. A city water bill charges \$0.00412 for each gallon used plus a \$22.40 service charge. Last month's bill was \$63.60. How many gallons were used?

Variable. $g = \text{the number of gallons used last month.}$

Equation. $0.00412g + 22.40 = 63.60$

$$0.00412g = 41.20$$

$$g = 10,000$$

Check: $0.00412(10,000) + 22.40 = 41.20 + 22.40 = 63.60 \checkmark$

Meaning. *10,000 gallons of water were used last month.*

This one already lands on a whole number, so nothing needs rounding. Not every “does it make sense” question ends in a change.

Questions 9 and 10 look ahead to multi-step equations, which arrive later this year. The unknown appears on both sides of the equal sign.

9. Nadia charges a \$20 booking fee plus \$12 an hour to babysit. Priya charges a \$50 booking fee plus \$6 an hour. At how many hours do the two charge the same amount?

Variable. $h = \text{the number of hours of babysitting.}$

Equation. $20 + 12h = 50 + 6h$

$$20 + 6h = 50$$

$$6h = 30$$

$$h = 5$$

Check: $20 + 12(5) = 80$, and $50 + 6(5) = 80 \checkmark$

Meaning. *At 5 hours both sitters charge \$80.*

Subtract 6h from both sides first. For jobs under 5 hours Nadia is cheaper; over 5 hours, Priya is.

10. Solve $5(x - 3) = 3x + 11$. Check your solution by substitution.

Equation. $5(x - 3) = 3x + 11$

$$5x - 15 = 3x + 11$$

$$2x - 15 = 11$$

$$2x = 26$$

$$x = 13$$

Check: $5(13 - 3) = 5(10) = 50$, and $3(13) + 11 = 39 + 11 = 50$ ✓

Meaning. $x = 13$ makes both sides equal 50.

Distribute first, then collect the x terms on one side. This is MA.8.AR.2.1 in its plainest form.